RetailIQ: Turning E-Commerce Transaction Data Into Strategic Business Intelligence

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***Abstract*—**Online stores generate massive amounts of daily sales data. However, raw sales records alone do not tell store owners how to make better business choices. This paper introduces RetailIQ, an end-to-end data analytics system built to clean, group, and analyze e-commerce transaction data from 2024–2025. Our framework combines data cleaning, basic exploratory analysis, RFM (Recency, Frequency, Monetary) customer grouping, market basket analysis for product pairs, sales forecasting, and an easy-to-use interactive dashboard built with Streamlit or Power BI. By connecting these tools into one clear pipeline, store managers can easily spot high-value customers, find items often bought together, and plan future inventory.

***Index Terms*—**E-commerce analytics, RFM segmentation, Market Basket Analysis, Sales Forecasting, Business Intelligence, Data Science Pipeline.

I. INTRODUCTION

EVERY time a customer buys something online, the store records timestamps, customer IDs, product types, prices, discounts, and payment methods. While businesses save all this information, raw records are messy and hard to read. Without a simple system to analyze the data, business owners miss out on clear patterns regarding customer loyalty and sales growth.

RetailIQ was created to solve this problem. Instead of looking at sales numbers or customer lists separately, RetailIQ connects data cleaning, customer grouping, item pairing, and future sales estimates into a single flow. The final results are displayed on a visual dashboard so team members can make quick, practical business choices.

II. PROBLEM STATEMENT

Turning raw online store records into clear business decisions usually runs into four main problems:

1) *Finding Loyal Customers:* It is hard to tell which customers spend the most and which ones are stopping their purchases.

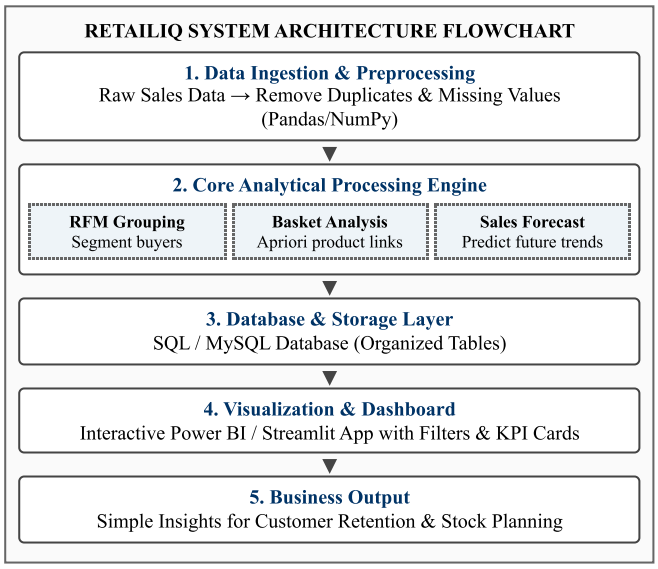
2) *Tracking Top Products:* Managers struggle to compare top sales items against slow-moving inventory across different regions.

3) *Spotting Combined Purchases:* It is difficult to see which products customers frequently buy together in the same order.

4) *Planning Future Sales:* Stores often rely on rough guesses rather than using past sales history to plan future stock.

III. SYSTEM ARCHITECTURE & SOLUTION

RetailIQ connects raw data processing all the way to a finished business dashboard in five main steps:



A. Data Preparation & Cleaning

First, the system loads transaction files and fixes broken data. It drops duplicate orders, fills in missing values, and turns text columns into proper numbers, dates, and prices using Python's Pandas library.

B. RFM Customer Grouping

To understand customer buying habits, RetailIQ uses RFM scoring:

- **Recency (R):** How recently a customer placed an order.
- **Frequency (F):** How many total orders they made.
- **Monetary (M):** Total money spent.

Using these three simple scores, customers are sorted into helpful groups like *Loyal Customers*, *New Buyers*, and *At-Risk Customers* who might need special discounts to return.

C. Market Basket Analysis (Item Pairings)

Using algorithms like Apriori or FP-Growth (via Python's `mlxtend` package), RetailIQ finds items that often land in the same shopping cart. For example, if customers who buy a phone frequently buy a screen protector too, store managers can bundle them together for quick cross-selling.

D. Sales Forecasting

The system uses historical sales numbers to spot monthly and seasonal trends. It fits basic regression models to predict upcoming sales so store owners do not run out of popular items.

IV. TOOLS & TECHNOLOGIES

The RetailIQ pipeline is built with standard, reliable open-source Python libraries and visual display tools:

TABLE I
PROJECT TOOLS & TECHNOLOGIES

Task / Purpose	Tools Used
Programming & Logic	Python (Jupyter Notebook)
Data Processing	Pandas, NumPy
Visual Charts	Matplotlib, Seaborn, Plotly
Machine Learning & Association	Scikit-learn, mlxtend (Apriori)
Interactive Dashboard	Power BI / Streamlit

V. IMPLEMENTATION PLAN & SUMMARY

The project implementation follows a 10-phase roadmap, connecting initial data intake directly to final business outputs:

TABLE II
ANALYTICAL MODULES & EXPECTED OUTPUTS

Analytics Step	Method	Business Benefit
Customer Grouping	RFM Scoring	Find loyal buyers & win back inactive ones
Basket Analysis	Apriori / FP-Growth	Create smart product bundles and discounts
Sales Forecast	Time-Series / Regression	Avoid stock shortages during peak seasons
Visual Dashboard	Streamlit / Power BI	Quick KPI views for store managers

VI. CONCLUSION & FUTURE WORK

RetailIQ gives online store managers a complete toolkit to convert raw sales data into actionable decisions. By combining cleaning, customer groups, product pairing, and forecasting into one dashboard, businesses gain clear insights into what customers buy and how to plan ahead. Future work will focus on testing real-time live data streams, improving forecasting models, and introducing automated email offers for at-risk customers.

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